

## **Predicting Digital Experiment Success Using A/B Testing Metrics**

### **Business Problem**

Digital experimentation and A/B testing are commonly used to improve website performance, increase conversions, and better understand user behavior. Organizations frequently run experiments on landing pages, checkout flows, and other digital experiences to identify changes that may improve business outcomes. However, many experiments fail to produce meaningful improvements, creating challenges when organizations attempt to prioritize testing efforts and allocate resources efficiently.

This project explores whether engagement and experimentation metrics can be used to predict experiment success using machine learning classification techniques.

### **Background / History**

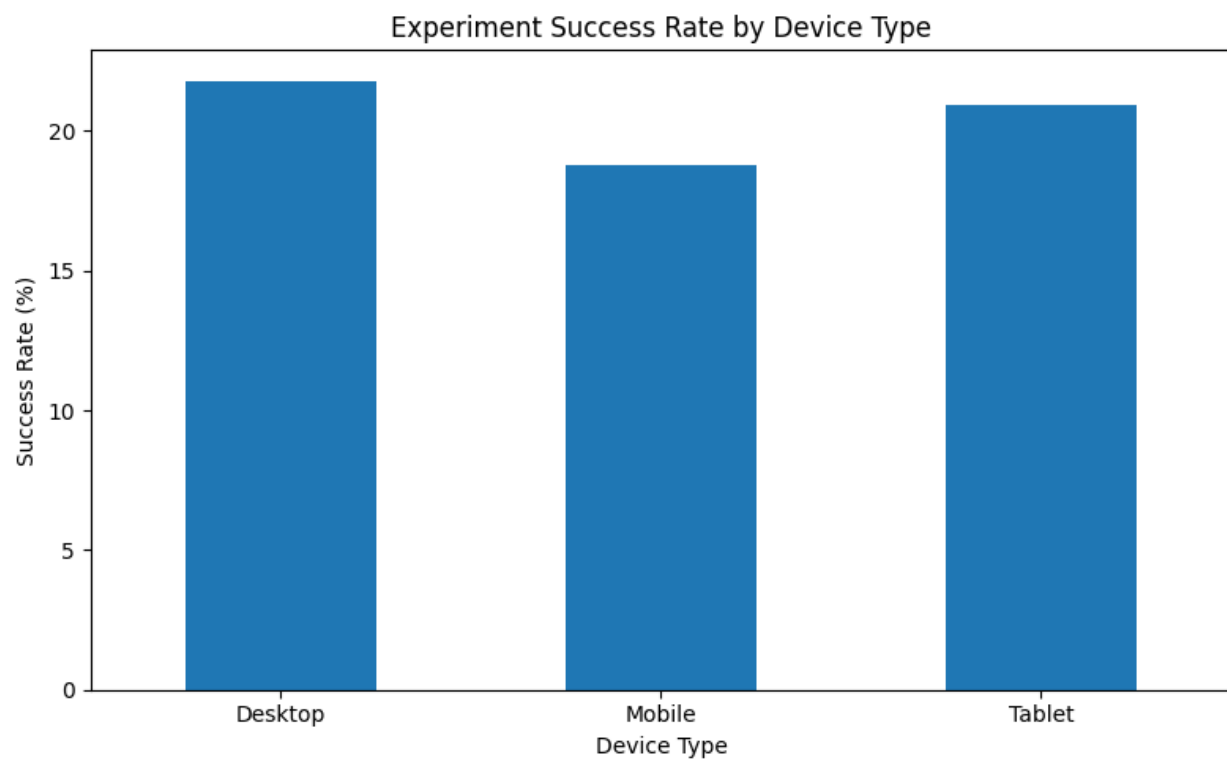
A/B testing has become an important part of digital optimization strategies across many industries (Kohavi et al., 2020). Organizations increasingly use experimentation platforms and behavioral analytics to improve engagement and conversion performance through data-driven decision-making.

### **Data Explanation**

This project used a synthetic dataset containing 1,500 simulated A/B testing experiments designed to represent common digital optimization scenarios. The dataset included variables such as device type, traffic volume, bounce rate, session duration, click-through rate, conversion metrics, lift percentage, p-values, and experiment outcome indicators.

Experiments were classified as successful only when the p-value was below 0.05 and the lift percentage exceeded 3%. During the modeling phase, target leakage was identified because variables directly connected to experiment outcomes were included in the feature set. After removing those variables, model performance decreased substantially and produced more realistic evaluation results.

**Figure 1. Dataset Variables and Descriptions**



## Methods

Exploratory data analysis was performed to evaluate experiment outcome patterns, metric distributions, traffic relationships, and correlations among variables.

Logistic Regression and Random Forest classification models were used to evaluate predictive performance. Before training, categorical variables were converted into numerical format using

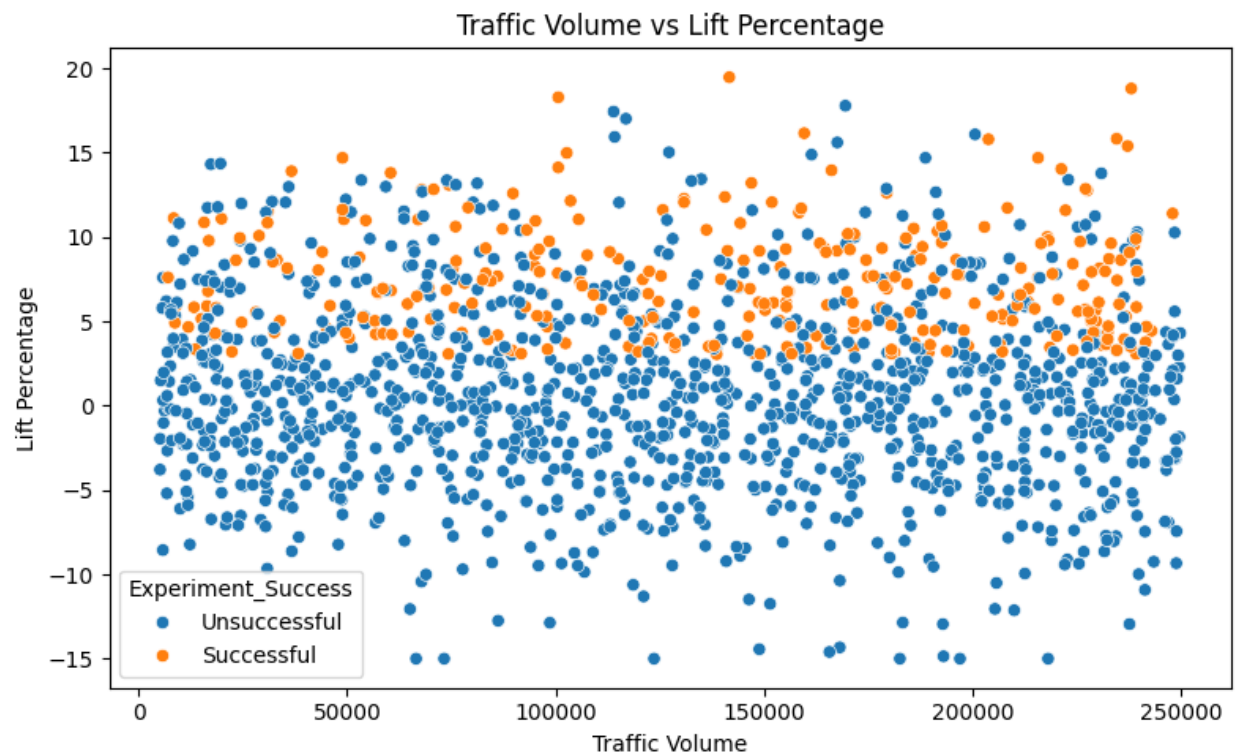
one-hot encoding, and the dataset was divided into training and testing subsets using an 80/20 split.

Model performance was evaluated using accuracy scores, classification reports, confusion matrices, and feature importance analysis.

## Analysis

The exploratory analysis revealed several patterns within the simulated experimentation data. Device-level analysis showed relatively small differences in experiment success rates across desktop, mobile, and tablet environments.

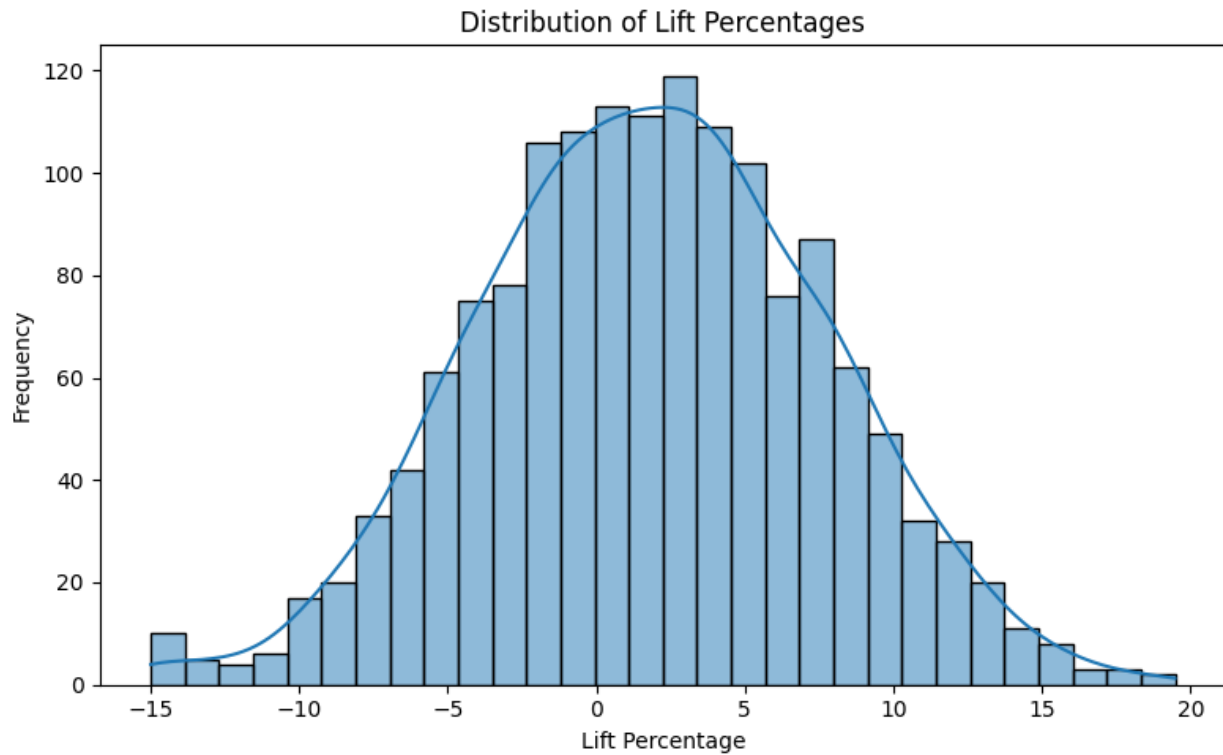
**Figure 2. Experiment Success Rate by Device Type**



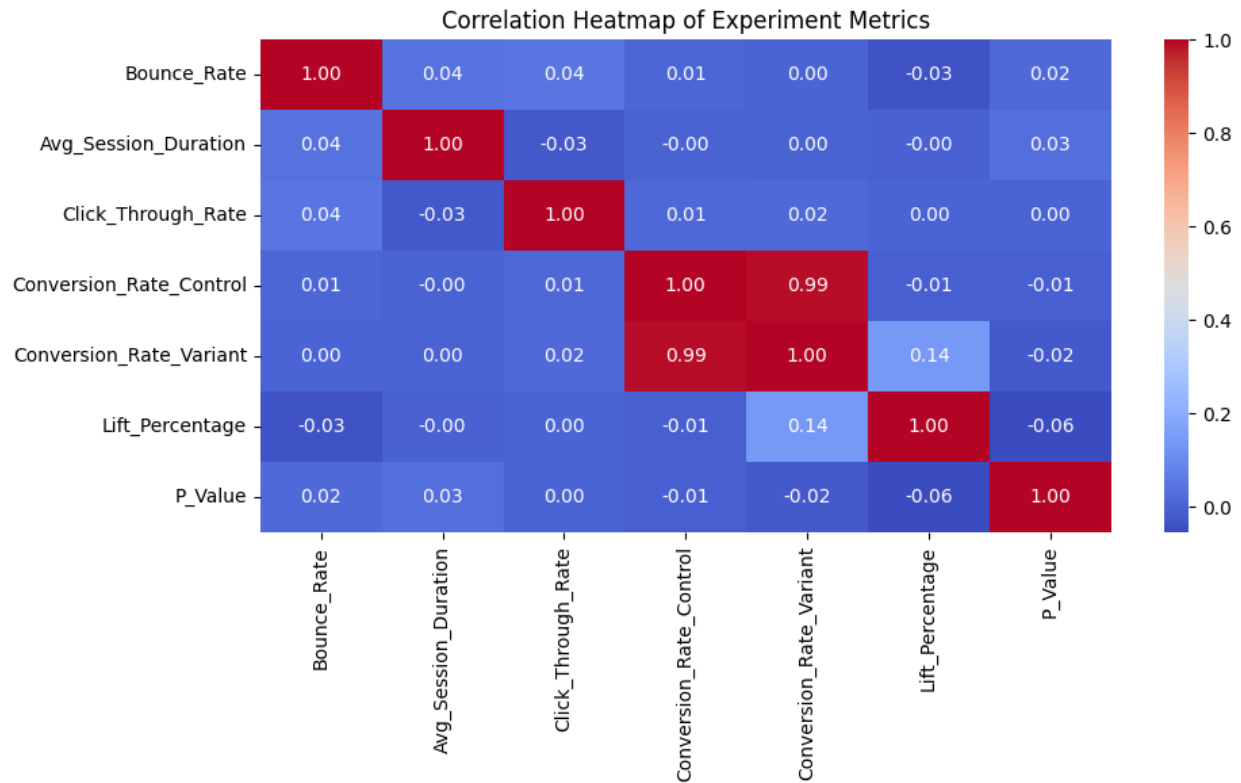
The relationship between traffic volume and lift percentage showed that higher traffic alone did not guarantee successful experimentation outcomes. Successful experiments appeared more

frequently within positive lift ranges, although several high-traffic experiments still produced minimal or negative performance improvements.

**Figure 3. Traffic Volume vs. Lift Percentage**



Correlation analysis revealed generally weak relationships among most variables, suggesting that experimentation outcomes were influenced by multiple interacting behavioral factors rather than a single dominant metric.

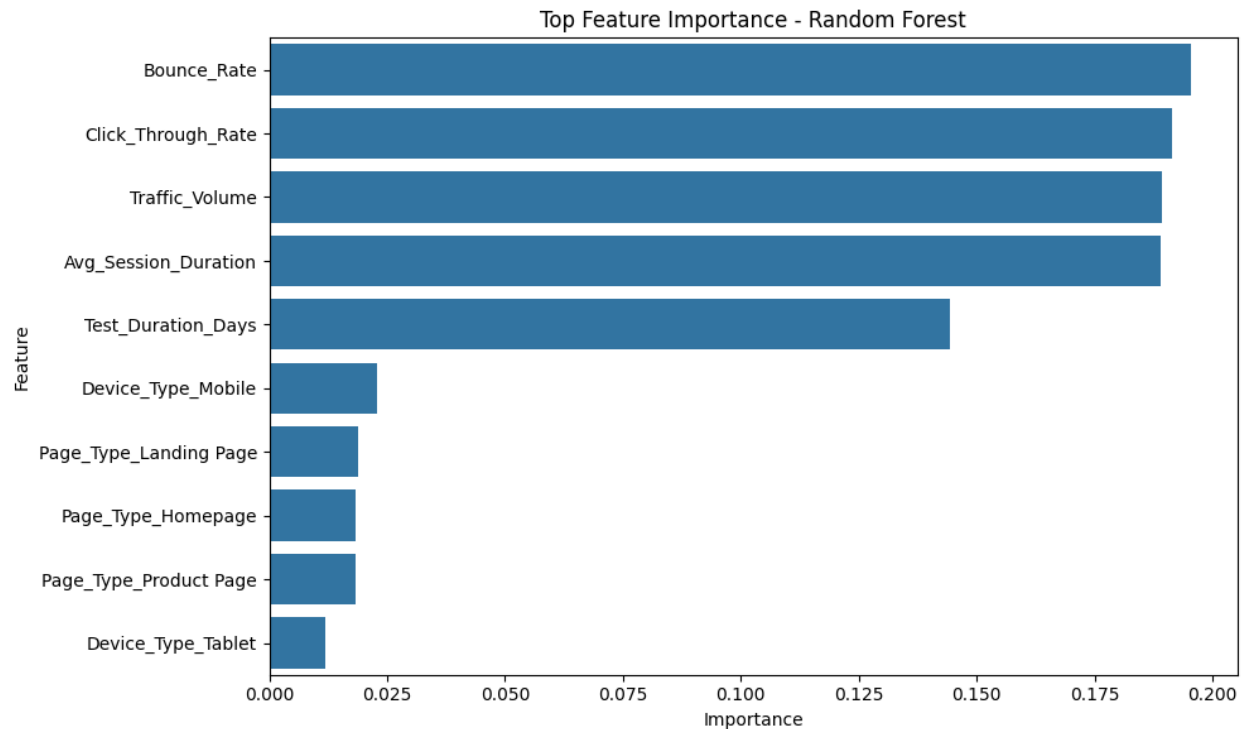
**Figure 4. Correlation Heatmap of Experiment Metrics**

One of the most important findings from the project involved target leakage during model development. Initial models achieved unrealistically high accuracy because variables directly tied to the target outcome were included within the feature set. After removing those variables, model performance decreased substantially and produced more realistic evaluation results.

Following leakage correction, the Logistic Regression model achieved an accuracy score near 79%, while the Random Forest model achieved approximately 78% accuracy. Although both models performed reasonably well overall, they struggled to consistently identify successful experiments.

Feature importance analysis from the Random Forest model showed that click-through rate, bounce rate, session duration, and traffic-related engagement metrics contributed most strongly to model behavior after leakage correction.

**Figure 5. Random Forest Feature Importance Analysis**



## Conclusion

This project demonstrated that machine learning techniques can provide analytical insight within digital experimentation environments, although predictive performance remained limited after realistic validation practices were applied.

The analysis also reinforced the importance of proper feature selection and realistic model evaluation practices when developing predictive analytics systems.

## **Assumptions**

This project assumed that the synthetic dataset reasonably reflected experimentation behavior commonly observed within digital optimization environments. It was also assumed that engagement metrics such as bounce rate, click-through rate, and session duration represented variables commonly monitored during A/B testing initiatives.

## **Limitations**

The primary limitation of this project involved the use of synthetic data rather than real-world experimentation data. Additionally, class imbalance reduced the models' ability to consistently identify successful experiments, resulting in very low recall scores for positive outcomes.

## **Challenges**

One of the largest challenges involved identifying and correcting target leakage during model development. Another challenge involved balancing realism and variability while generating synthetic experimentation data.

## **Future Uses / Additional Applications**

Future versions of this project could incorporate real-world experimentation datasets, additional behavioral variables, and more advanced machine learning approaches such as gradient boosting models.

## **Recommendations**

Organizations using predictive experimentation analytics should carefully evaluate feature selection processes and prioritize behavioral engagement metrics when evaluating experiment effectiveness.

## Implementation Plan

A real-world implementation would require centralized experimentation data collection, ongoing model validation, and analyst oversight to ensure responsible predictive model usage.

## Ethical Assessment

Organizations implementing predictive experimentation analytics must ensure responsible data collection practices and avoid overreliance on automated decision-making. This project also demonstrated how target leakage can produce misleading model performance if predictive systems are not evaluated responsibly.

## References

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## Appendix

### Variables Included in the Synthetic Dataset

| <u>Variable</u>                | <u>Description</u>                                                           |
|--------------------------------|------------------------------------------------------------------------------|
| <b>Experiment_ID</b>           | Unique identifier assigned to each simulated experiment                      |
| <b>Device_Type</b>             | Device category associated with experiment traffic (Desktop, Mobile, Tablet) |
| <b>Page_Type</b>               | Type of webpage involved in the experiment                                   |
| <b>Traffic_Volume</b>          | Number of users exposed to the experiment                                    |
| <b>Test_Duration_Days</b>      | Length of time the experiment was active                                     |
| <b>Bounce_Rate</b>             | Percentage of users leaving the page without further interaction             |
| <b>Avg_Session_Duration</b>    | Average session duration measured in seconds                                 |
| <b>Click_Through_Rate</b>      | Percentage of users clicking targeted content                                |
| <b>Conversion_Rate_Control</b> | Conversion rate for the control experience                                   |
| <b>Conversion_Rate_Variant</b> | Conversion rate for the experiment variation                                 |
| <b>Lift_Percentage</b>         | Percent difference between control and variation conversion rates            |
| <b>P_Value</b>                 | Statistical significance measurement                                         |

|                                  |                                                                             |
|----------------------------------|-----------------------------------------------------------------------------|
| <b>Statistically_Significant</b> | Indicates whether the experiment met the statistical significance threshold |
| <b>Experiment_Success</b>        | Final classification outcome for the experiment                             |